Venkata Lakshmi Parimala Pasupuleti

Machine Learning Software Engineer · Python · PyTorch · HuggingFace · Eval-Driven Development · Distributed Systems

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Machine Learning Software Engineer — Apple Watch · Apple

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Dear Hiring Team,

The features this team ships — DoubleTap, AssistiveTouch, Handwashing — represent what I find most compelling about applied ML: not models running in a cloud cluster, but intelligence embedded in a device that fits on a wrist, operating under real power and latency constraints, detecting meaningful human behavior in noisy real-world conditions. That problem — moving from a promising model to a shipped, reliable, user-facing feature — is what I want to spend my career on.

My ML depth comes from doing the full cycle, not just the training step. At GMU, I ran data collection, feature engineering, model training (HuggingFace Transformers + PyTorch), quantitative evaluation, and inference API deployment — then stood in front of faculty and defended the methodology. I built AgentProbe specifically because I needed a rigorous way to measure whether my AI systems were actually improving across iterations: CI-integrated regression harness, 5-metric eval engine, LLM-as-judge failure analysis, golden datasets. When I evaluated LeaseI — an 8-agent platform I built from scratch — I didn't eyeball outputs; I built a pass@k harness with multi-trial sampling and a regression gate that had to pass before anything shipped. 93.6% precision, 87.9% F1, pass@3 88.3% across 1,240 test cases. That methodology — define the metric, build the harness, gate on it — is how I approach any ML problem.

The production engineering side is eight years deep. At Entrata I owned distributed systems serving 100K+ monthly active users at 99.5% uptime, profiled and optimized 200+ SQL queries for a 30% performance gain, and rebuilt microservices in Docker for a 50% scalability improvement. I understand what it means to ship something that can't break, not just something that demos well. I'm an active Swift learner and I know the Apple platform fundamentals are the steepest part of the ramp — but the ML rigor, the production engineering discipline, and the instinct for shipping over demoing are already there.

I'd welcome the chance to talk about the work and what I'd bring to this team.

Sincerely,

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